CNC PROGRAM SHEET

Part: MOTOR_MOUNT | Rev: A | Date: 2026-06-22

Programmer: CNC-AI

Material: ALUMINIUM

Max RPM: 4500 Coolant: through spindle

Setups: 3

Part envelope: 55.3 x 104.8 x 21.5 mm

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face centre of part	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	CD_60_2MM	60-degree combined center drill, 2mm tip, 5mm body	2.0	Sandvik Coromant	H10F	spot_drill
T02	EM_2FL_2MM	2-flute solid carbide end mill 2mm â€” aluminium geometry	2.0	Sandvik Coromant	1630	contour_mill
T03	DRILL_2.5MM	Solid carbide jobber drill 2.5mm	2.5	Sandvik Coromant	H10F	twist_drill
T04	DRILL_3.2MM	Solid carbide jobber drill 3.2mm â€” standard size for Ã˜3.2mm precision holes	3.2	Sandvik Coromant	H10F	twist_drill
T05	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T06	BALL_2FL_AL_4MM	2-flute solid carbide ball-nose end mill 4mm â€” aluminium geometry	4.0	Sandvik Coromant	1630	profile_3d_contour
T07	DRILL_5.0MM	Solid carbide jobber drill 5.0mm	5.0	Sandvik Coromant	H10F	twist_drill
T08	SPOT_090_6MM	90-degree solid carbide spot drill, 6mm shank	6.0	Sandvik Coromant	H10F	spot_drill
T09	EM_2FL_6MM	2-flute solid carbide end mill 6mm â€” aluminium geometry	6.0	Sandvik Coromant	1630	pocket_mill
T10	DRILL_10.0MM	Solid carbide jobber drill 10.0mm â€” also used as pilot drill	10.0	Sandvik Coromant	H10F	pilot_drill, twist_drill
T11	EM_10MM_2FL	--	10.0	--	--	pocket_mill
T12	DRILL_16.0MM	Solid carbide jobber drill 16.0mm â€” also used as core drill	16.0	Sandvik Coromant	H10F	core_drill
T13	FACEMILL_40MM	40mm indexable face mill â€” 6 inserts	40.0	Sandvik Coromant		face_mill

SETUP 1 -- -Z approach (rear face)

PRINCIPAL | 30 clusters | 45 operations

Fixture	90° angle plate or tombstone fixture.
Description	Rear-face setup. Rotate part 90° so rear face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=CENTER (X=0.000mm) Y=CENTER (Y=-20.711mm) Z=TOP
Face (W x H)	55.3 x 104.8 mm
Max depth	21.5 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
2	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
3	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
4	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
5	ALU 40 FACEMILL FACE FINISH	T13	40.0	0	0.29	4500	3240	0.294	0.0	0.0	00:02
6	ALU 40 FACEMILL FACE FINISH	T13	40.0	0	0.29	4500	3240	0.294	0.0	0.0	00:02
7	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
8	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
9	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
10	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
11	ALU CENTER DRILL	T08	6.0	0	3.0	420	21	3.0	--	--	00:09
12	ALU 10 DRILL	T10	10.0	0	1.5	4500	495	1.5	--	--	00:06
13	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
14	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
15	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
16	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
17	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
18	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
19	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
20	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
21	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
22	ALU 3.2 DRILL	T04	3.2	0	1.5	1190	80	1.5	--	--	00:12
23	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
24	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
25	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
26	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
27	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
28	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:04
29	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
30	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:04
31	ALU 2 ENDMILL BOSS RF	T02	2.0	0	0.01	4500	90	0.009	0.1	0.1	00:06
32	ALU 2 ENDMILL BOSS FINISH	T02	2.0	0	0.01	4500	90	0.009	0.0	0.0	00:06
33	ALU 2 ENDMILL BOSS RF	T02	2.0	0	0.02	4500	90	0.019	0.1	0.1	00:06
34	ALU 2 ENDMILL BOSS FINISH	T02	2.0	0	0.02	4500	90	0.019	0.0	0.0	00:06
35	ALU 40 FACEMILL FACE FINISH	T13	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05
36	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.38	3580	358	0.375	0.0	0.0	01:11
37	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.38	3580	358	0.375	0.0	0.0	01:11
38	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.81	3580	358	0.4	0.0	0.0	00:51
39	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.81	3580	358	0.4	0.0	0.0	00:51
40	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.38	3580	358	0.375	0.0	0.0	01:11
41	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.38	3580	358	0.375	0.0	0.0	01:11
42	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.62	3580	358	0.4	0.0	0.0	00:51
43	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.62	3580	358	0.4	0.0	0.0	00:51
44	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.88	3580	358	0.4	0.0	0.0	00:51
45	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	0.88	3580	358	0.4	0.0	0.0	00:51
46	ALU 10 ENDMILL STOCK ROUGHING RF	T11	10.0	0	--	3980	1500	14.0	--	--	11:00

ESTIMATED SETUP TIME: 24:58 (includes tool changes)

SETUP 2 -- -X approach (left side face)

PRINCIPAL | 9 clusters | 9 operations

Fixture	90° angle plate or tombstone fixture.
Description	Left-side face setup. Rotate part 90° so left face is accessible. Clamp on bottom and right faces.
Rotation	None -- standard orientation
WCS	G55
Origin X/Y/Z	X=CENTER (Y=-20.711mm) Y=CENTER (Z=2.750mm) Z=TOP
Face (W x H)	104.8 x 21.5 mm
Max depth	55.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T05	4.0	0	2.0	640	32	2.0	--	--	00:04
2	ALU 5 DRILL	T07	5.0	0	0.74	1400	98	0.738	--	--	00:15
3	ALU 40 FACEMILL FACE FINISH	T13	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05
4	ALU 40 FACEMILL FACE FINISH	T13	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05
5	ALU 40 FACEMILL FACE FINISH	T13	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05
6	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	3.57	3580	358	0.4	0.0	0.0	00:51
7	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	3.0	3580	358	0.4	0.0	0.0	00:51
8	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	4.75	3580	358	0.4	0.0	0.0	00:51
9	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	1.69	3580	358	0.4	0.0	0.0	00:51
10	ALU 10 ENDMILL STOCK ROUGHING RF	T11	10.0	0	--	3980	1500	14.0	--	--	13:26

ESTIMATED SETUP TIME: 18:06 (includes tool changes)

SETUP 3 -- +Y approach (top — default VMC position)

PRINCIPAL | 10 clusters | 13 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G56
Origin X/Y/Z	X=CENTER (X=0.000mm, 27.67mm from xmin edge) Y=CENTER (Z=2.750mm, 10.76mm from zmin edge) Z=TOP
Face (W x H)	55.3 x 21.5 mm
Max depth	104.8 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T08	6.0	0	3.0	420	21	3.0	--	--	00:09
2	ALU 10 DRILL	T10	10.0	0	17.03	4500	540	17.033	--	--	00:02
3	ALU 16 DRILL	T12	16.0	0	17.03	3580	573	17.033	--	--	00:02
4	ALU 6 ENDMILL POCKET RF	T09	6.0	0	--	3020	483	6.0	0.1	0.1	00:05
5	ALU 6 ENDMILL POCKET FINISH	T09	6.0	0	--	3020	483	6.0	0.0	0.0	00:05
6	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.99	3580	358	0.4	0.0	0.0	01:31
7	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.4	3580	358	0.4	0.0	0.0	01:31
8	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.4	3580	358	0.4	0.0	0.0	01:31
9	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.99	3580	358	0.4	0.0	0.0	01:31
10	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:51
11	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:51
12	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.62	3580	358	0.4	0.0	0.0	01:41
13	ALU 4 BALLNOSE FILLET FINISH	T06	4.0	0	9.62	3580	358	0.4	0.0	0.0	01:41

ESTIMATED SETUP TIME: 12:17 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
SUBSTITUTION	C2	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C3	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
RPM CAPPED	C7	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
WARNING	C22	spot_drill	WARNING: no spot drill covers dia=16.1mm, using largest available 6.0mm Field-validated: RPM ~425 F21 â€” slow spot for p
SUBSTITUTION	C22	pilot_drill	SUBSTITUTION: 9.66mm is non-standard. Using nearest standard size 10.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C22	pilot_drill	Effective Vc = 141.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C22	core_drill	SUBSTITUTION: 16.1mm is non-standard. Using nearest standard size 16.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C25	twist_drill	Effective Vc = 141.4 m/min (reduced from 220 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C31	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C32	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C33	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C34	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
RPM CAPPED	C36	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C36	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C65	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C67	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C68	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C69	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.